

We are given two functions

$$f(x) := \cos(2\pi x), \quad x \in [0, 1] \quad \text{and} \quad g(x) := e^{3x} \sin(2x), \quad x \in [0, \pi/4]$$

We have that the interpolation error is given as

$$y(x) - p_n(x) = \frac{y^{(n+1)}(\xi)}{(n+1)!} \pi_{n+1}(x), \quad \text{where } \pi_{n+1}(x) := \prod_{i=0}^n (x - x_i) \quad \text{and} \quad \xi = \xi(x) \in [a, b]$$

which gives

$$\|y(x) - p_n(x)\|_\infty \leq \frac{M_{n+1}}{(n+1)!} \|\pi_{n+1}(x)\|_\infty, \quad \text{where } M_{n+1} := \max_{\xi \in [a, b]} |y^{(n+1)}(\xi)|$$

$f \in C^\infty[a, b]$ is not sufficient for convergence, as seen with Runge's function interpolated with equidistant nodes on $[-5, 5]$.

In order for the interpolation polynomial to converge, M_{n+1} needs to grow slower than $\|\pi_{n+1}(x)\|$ grows for $n \rightarrow \infty$.

For the equidistant nodes on $[a, b]$ we can claim that

$$h = (b - a)/n \implies \|\pi_{n+1}(x)\| \leq \frac{1}{4} n! h^{n+1},$$

and the proof for this bound follows:

Fix $x \in [a, b]$ and $x \in [x_j, x_{j+1}]$ for $0 \leq j \leq n-1$. With $t = x - x_j \in [0, h]$ the two nearest factors give $|x - x_j| |x - x_{j+1}| = t(h - t) \leq h^2/4$, since $t = h/2$ is the maximum.

All other nodes lie outside $[x_j, x_{j+1}]$, so their distance to x is at most the distance to the far endpoint of that subinterval. $|x - x_i| \leq (j+1-i)h$ for $i \leq j-1$, and $|x - x_i| \leq (i-j)h$ for $i \geq j+2$. Multiplying these $n-1$ factors gives $h^j (j+1)! \cdot h^{n-j-1} (n-j)! = h^{n-1} (j+1)! (n-j)!$, where empty products are 1. Hence $|\pi_{n+1}(x)| \leq \frac{h^{n+1}}{4} (j+1)! (n-j)!$.

Finally, with $k = j+1 \in \{1, \dots, n\}$ we have $(j+1)! (n-j)! = k! (n+1-k)! = (n+1)! / \binom{n+1}{k}$, and since binomial coefficients in a row are smallest at the ends, $\binom{n+1}{k} \geq n+1$. So $(j+1)! (n-j)! \leq n!$, and

$$|\pi_{n+1}(x)| \leq \frac{1}{4} n! h^{n+1}. \quad \blacksquare$$

And for Chebyshev nodes on $[a, b]$

$$\|\pi_{n+1}(x)\| = 2((b-a)/4)^{n+1},$$

which has the following proof:

Let $x_i = \frac{a+b}{2} + \frac{b-a}{2} \cos\left(\frac{(2i+1)\pi}{2n+2}\right)$, $i = 0, \dots, n$, be the Chebyshev nodes on $[a, b]$, and $\pi_{n+1}(x) = \prod_{i=0}^n (x - x_i)$.

Let then $d_i = \cos\left(\frac{(2i+1)\pi}{2n+2}\right)$ to the x_i . This gives $x - x_i = \frac{b-a}{2}(d - d_i)$, so $\pi_{n+1}(x) = \left(\frac{b-a}{2}\right)^{n+1} \prod_{i=0}^n (d - d_i)$.

We know that the d_i are the roots of the Chebyshev polynomial $T_{n+1}(d) = \cos((n+1) \arccos d)$, with leading coefficients 2^n . For such a monic polynomial, we can write

$$\prod_{i=0}^n (d - d_i) = 2^{-n} T_{n+1}(d),$$

Since T_{n+1} is a cosine function, it never exceeds 1 in absolute value, and $T_{n+1}(1) = 1$, so the maximum of the product on $[-1, 1]$ is exactly 2^{-n} .

Therefore

$$\|\pi_{n+1}\|_\infty = \left(\frac{b-a}{2}\right)^{n+1} 2^{-n} = \frac{(b-a)^{n+1}}{2^{2n+1}} = 2 \left(\frac{b-a}{4}\right)^{n+1}. \quad \blacksquare$$

By theorem 8.7 in *An introduction to Numerical Analysis* (Süli and Mayers) this is the smallest max norm any monic polynomial of degree $n+1$ can have on $[a, b]$, and this is therefore the best choice of nodes.

For the factor M_n we can show that for f it follows that

$$f^{(k)}(x) = (2\pi)^k \cos(2\pi x + k\pi/2), \quad k \geq 0,$$

which follows by induction: the formula holds for $k=0$, and differentiating the right-hand side gives $-(2\pi)^{k+1} \sin(2\pi x + k\pi/2) = (2\pi)^{k+1} \cos(2\pi x + (k+1)\pi/2)$. Hence

$$|f^{(n+1)}(x)| \leq (2\pi)^{n+1} =: M_{n+1},$$

and since the cosine is bounded by ± 1 on $[0, 1]$ for every n , $M_{n+1} = \max_{[0,1]} |f^{(n+1)}|$.

For g we similarly find that for $g(x) = \text{Im}(e^{(3+2i)x})$, we get

$$g^{(k)}(x) = \text{Im}\left((3+2i)^k e^{(3+2i)x}\right), \quad k \geq 0.$$

Therefore, on $[0, \pi/4]$,

$$|g^{(n+1)}(x)| \leq |3+2i|^{n+1} \left|e^{(3+2i)x}\right| = 13^{(n+1)/2} e^{3x} \leq 13^{(n+1)/2} e^{3\pi/4} =: M_{n+1}.$$

Here both $|\text{Im}(z)| \leq |z|$ and $e^{3x} \leq e^{3\pi/4}$ are used, and the two are not sharp at the same point. See the discussion of tightness below.

We find that $M_{n+1} \leq K \cdot C^{n+1}$, where K and C are positive constants independent of n , so the interpolation polynomial converges as $n \rightarrow \infty$.

This gives for equidistant nodes in max norm

$$\|f - p_n\|_\infty \leq \frac{(2\pi)^{n+1}}{(n+1)!} \cdot \frac{1}{4} n! h^{n+1} = \frac{1}{4(n+1)} (2\pi h)^{n+1} = \frac{1}{4(n+1)} \left(\frac{2\pi(b-a)}{n}\right)^{n+1},$$

$$\|g - p_n\|_\infty \leq \frac{13^{(n+1)/2} e^{3\pi/4}}{(n+1)!} \cdot \frac{1}{4} n! h^{n+1} = \frac{e^{3\pi/4}}{4(n+1)} (13^{1/2} h)^{n+1} = \frac{e^{3\pi/4}}{4(n+1)} \left(\frac{13^{1/2}(b-a)}{n}\right)^{n+1}.$$

For the Chebyshev nodes in max norm we find

$$\|f - p_n\|_\infty \leq \frac{(2\pi)^{n+1}}{(n+1)!} \cdot 2((b-a)/4)^{n+1} = \frac{2}{(n+1)!} \left(\frac{\pi(b-a)}{2}\right)^{n+1} = \frac{2}{(n+1)!} \left(\frac{\pi}{2}\right)^{n+1},$$

$$\|g - p_n\|_\infty \leq \frac{13^{(n+1)/2} e^{3\pi/4}}{(n+1)!} \cdot 2((b-a)/4)^{n+1} = \frac{2e^{3\pi/4}}{(n+1)!} \left(\frac{13^{1/2}(b-a)}{4}\right)^{n+1} = \frac{2e^{3\pi/4}}{(n+1)!} \left(\frac{13^{1/2}\pi}{16}\right)^{n+1}.$$

For equidistant nodes $h = (b-a)/n$, so

$$\frac{M_{n+1}}{(n+1)!} \|\pi_{n+1}\|_\infty \leq \frac{KC^{n+1}}{(n+1)!} \cdot \frac{1}{4} n! h^{n+1} = \frac{K}{4(n+1)} \left(\frac{C(b-a)}{n}\right)^{n+1} \rightarrow 0,$$

since $C(b-a)/n \leq 1/2$ for $n \geq 2C(b-a)$. The factor $n!$ cancels against $(n+1)!$, so the convergence comes from $h \rightarrow 0$. For Chebyshev nodes

$$\frac{M_{n+1}}{(n+1)!} \|\pi_{n+1}\|_\infty \leq \frac{KC^{n+1}}{(n+1)!} \cdot 2 \left(\frac{b-a}{4} \right)^{n+1} = \frac{2K}{(n+1)!} \left(\frac{C(b-a)}{4} \right)^{n+1} \rightarrow 0,$$

since $r^m/m! \rightarrow 0$ for every fixed $r > 0$.

For Runge's function $r(x) := 1/(1+x^2)$ on $[-5, 5]$ the geometric series gives

$$r(x) = \sum_{k=0}^{\infty} (-1)^k x^{2k}, \quad |x| < 1 \quad \implies \quad r^{(2k)}(0) = (-1)^k (2k)!,$$

so for odd n we have $M_{n+1} \geq |r^{(n+1)}(0)| = (n+1)!$, and the factorial in the error formula disappears. Evaluating π_{n+1} at the midpoint of the first subinterval, $x^* := a + h/2$ with $h = 10/n$,

$$\|\pi_{n+1}\|_\infty \geq |\pi_{n+1}(x^*)| = h^{n+1} \prod_{i=0}^n \left| \frac{1}{2} - i \right| = \frac{(2n-1)!!}{2^{n+1}} h^{n+1} \sim \frac{10}{\sqrt{2}n} \left(\frac{10}{e} \right)^n \quad [\text{Stirling's approx}],$$

so that

$$\frac{M_{n+1}}{(n+1)!} \|\pi_{n+1}\|_\infty \geq \|\pi_{n+1}\|_\infty \rightarrow \infty, \quad n \rightarrow \infty.$$

Here M_{n+1} grows factorially rather than geometrically, which is exactly what separates r from f and g .

We use the ML inequality to find a relationship between 2 norm and the max norm:

$$\|f - p_n\|_2^2 = \int_a^b (f - p_n)^2 dx \stackrel{ML}{\leq} (b-a) \max_{x \in [a,b]} (f - p_n)^2$$

$$\implies \|f - p_n\|_2 \leq \sqrt{b-a} \max_{x \in [a,b]} |f - p_n| = \sqrt{b-a} \|f - p_n\|_\infty$$

And then we use the approximations to find a relationship in the discrete case, using the above.

How tight is the bound?

The factor $\max_{[a,b]} |f^{(n+1)}|$ does not depend on the choice of nodes, so π_{n+1} is what varies. For the Chebyshev nodes

$$\prod_{i=0}^n (x - x_i) = \left(\frac{b-a}{2} \right)^{n+1} 2^{-n} T_{n+1}(d), \quad d = \frac{2x - a - b}{b - a},$$

and T_{n+1} has the values ± 1 at $n+2$ points of $[-1, 1]$, so the product oscillates between $\pm 2((b-a)/4)^{n+1}$ across the whole interval. For equidistant nodes the product is by contrast extremely small in the middle of the interval and largest in the two outermost subintervals.

The reason there is a gap is that in the error formula for the interpolation polynomial there is the factor

$$f^{(n+1)}(\xi(x)) \prod_i (x - x_i),$$

and the bound uses the maximum of both the factors, but they don't have to have their maxima at the same points.

The Chebyshev nodes minimize $\max_x |\prod_i (x - x_i)|$, and one can compare the factor $|\prod_i (x - x_i)|$ for equidistant and Chebyshev nodes and see that

$$\begin{aligned} \text{Equidistant: } \quad & \left| \prod_{\text{eq}} \right| \leq n! h^{n+1} \sim \sqrt{2\pi/n} e^{-n} (b-a)^{n+1} \quad [\text{Stirling's approx}] \\ \text{Chebyshev: } \quad & \left| \prod_{\text{cheb}} \right| \leq 2((b-a)/4)^{n+1} \end{aligned}$$

$$\Rightarrow \quad \frac{|\prod|_{\text{eq}}}{|\prod|_{\text{cheb}}} \sim (4/e)^n \simeq 1.47^n$$

What the comparison shows is that the node product for equidistant nodes exceeds the Chebyshev one by a factor growing like 1.47^n .